

YEAR 9 (AFTER PREP) · THEME 7 · LIFE IN NATURE & GLOBAL PROBLEMS

Reading

B2.1

When the Ground Shakes: The Real Cost of a Natural Disaster · Can Technology Predict the Next Disaster?

Hazırlık Sonrası 9.
Sınıf

Name – Surname

Class

No

Date

Aim Understand the main idea and details, make inferences and connect the text to your own life.

1 Before you read

Make predictions before you read.

1. Look at the title. What do you expect this text to compare or describe?
2. What do you already know about the effects of an earthquake on a town or a natural environment?

When the Ground Shakes: The Real Cost of a Natural Disaster

- 1 At 4:17 in the morning, the small town of Elmalı, which had never experienced a serious earthquake before, began to shake violently for almost forty seconds. Buildings that had stood for decades collapsed within minutes, and the streets, which were usually silent at that hour, filled with dust, noise and frightened residents searching for their families. By sunrise, rescue teams whose job it was to search the rubble had already pulled dozens of survivors from collapsed buildings, but for many families the search continued for days.
- 2 The disaster's impact on the surrounding countryside was just as severe as its impact on the town itself. The earthquake triggered several landslides that blocked mountain roads and changed the course of a small river, flooding farmland that had fed the community for generations. Forests on the hillsides, whose roots normally hold the soil in place, were badly damaged, which made the land even more vulnerable to future landslides. Local wildlife, including birds and small mammals that depended on those forests, lost much of their natural habitat almost overnight.
- 3 In the weeks that followed, thousands of people who had lost their homes were moved to temporary shelters, and this displacement created its own challenges: overcrowding, limited clean water, and the stress of an uncertain future. Yet the relief effort that followed also showed remarkable resilience. Volunteers whose only connection to Elmalı was a desire to help arrived from across the country, and within a month a temporary school, which the community built almost entirely by hand, was already open for local children.
- 4 Experts who study natural disasters point out that Elmalı's experience is far from unique. Every year, similar events devastate communities that are simply unprepared for what nature can do — a

reminder that understanding both the human and the environmental cost of a disaster is the first step towards being ready for the next one.

313 words

2 While you read

5 × 2 pts

Circle the correct option.

- What is the main idea of the text?
 - Elmalı's earthquake only damaged buildings, not the countryside.
 - The earthquake had a serious impact on both the town and the natural environment around it.
 - The relief effort in Elmalı failed completely.
 - The earthquake triggered several _____ that blocked mountain roads and changed the course of a river.
 - According to the text, why did the damaged forests make the land more vulnerable to future landslides?
 - Because the trees attracted more rain
 - Because their roots normally hold the soil in place
 - Because the wildlife left the area
 - The temporary school in Elmalı was built entirely by government workers. ☐ Correct ☐ Incorrect
- _____
- Thousands of people who had lost their homes experienced _____, which created overcrowding and limited clean water.

3 After you read

Write short answers. Then compare with a partner.

- Which effect of the earthquake — on people or on nature — do you think was more serious? Why?

- Write one sentence describing how you think you would feel in Elmalı's situation, using a relative clause.

1 Before you read

Make predictions before you read.

- What do you think technology can and cannot do to prevent damage from a natural disaster?

- Have you ever heard of an app or a system that warns people about floods, earthquakes or wildfires?

Can Technology Predict the Next Disaster?

- 1 For most of human history, natural disasters arrived almost without warning. Today, however, artificial intelligence and data analysis are changing that completely. Scientists who study climate patterns now use algorithms that can process huge amounts of satellite data, weather records and historical information — far more than any human team could analyse alone — to identify the early signs of an earthquake, flood or wildfire before it happens.
- 2 One clear example is the early warning system that many coastal cities have started to install. Sensors, which are placed along riverbanks and coastlines, constantly send data to a central system whose algorithm compares current conditions with thousands of past events. When the pattern matches a known danger, the system can send an alert to residents' phones minutes — or sometimes hours — before a flood or a tsunami arrives. In regions prone to wildfires, similar technology uses satellite images to detect the first signs of smoke, often long before anyone on the ground notices anything unusual.
- 3 The benefits of this technology are not limited to prediction. After a disaster strikes, hospitals whose emergency departments would otherwise be overwhelmed can use data models to estimate how many patients they are likely to receive and what kind of injuries to expect, allowing them to prepare staff and supplies more effectively. Humanitarian organisations, which used to rely mainly on phone calls and paper reports, now use real-time data to decide where relief teams are needed most urgently.
- 4 Of course, technology is not a perfect solution. Systems that depend on expensive sensors and reliable internet connections are often unavailable in the poorest and most vulnerable communities — the very places that frequently need them most. Experts agree that AI and data can dramatically reduce the damage a disaster causes, but only if the people who design these systems remember to make them available to everyone, not just to wealthy cities that can already afford them.

320 words

2 While you read

5 × 2 pts

Circle the correct option.

1. What is the main purpose of the text?
 - a To explain why technology cannot help with natural disasters
 - b To describe how AI and data are changing disaster prediction and response
 - c To list every country that has an early warning system
2. Sensors placed along riverbanks and coastlines send data to a central system whose _____ compares current conditions with past events.
3. How can data models help hospitals after a disaster?
 - a By replacing doctors completely
 - b By estimating patient numbers and likely injuries so staff can prepare
 - c By predicting exactly which patients will survive
4. According to the text, early warning technology is equally available to every community in the world.
☐ Correct ☐ Incorrect
5. Humanitarian organisations now use _____ data to decide where relief teams are needed most urgently.

3 After you read*Write short answers. Then compare with a partner.*

1. Do you agree that AI can 'dramatically reduce the damage a disaster causes'? Why or why not?

2. What is one problem with relying on technology for disaster prediction, according to the text?

